Database Systems Project

OnePharmacy Database

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Abstract::

The development of a comprehensive pharmacy administration database managed by the government that addresses two significant issues: the spread of phony medications and evaluating straightforwardness. While current frameworks regularly focus on overseeing solutions or stock, this drive widens the degree to address public worries about questionable medication providers and secret markups.

Fundamental Highlights:

Registration and aggregation of data for pharmacies: All enlisted drug stores in the designated region should enroll and submit significant information, for example, their location, telephone number, permit history, and stock of prescriptions. By get-together such a lot of information, a focal storehouse for exact drug store data is made.

Drug stores will be expected to present their selling costs for each medication along with their connected buy costs under the provisions of normalized evaluating and markup hailing. The device will automatically compute and display the results. A configurable edge will set off a caution at whatever point a drug store's markup surpasses a predefined limit, possibly inordinate exploitative to the two controllers and purchasers.

Drug Serialization and Confirmation: A pivotal element will be the incorporation of a prescription serialization framework. Every unit of medicine will be relegated to an extraordinary identifier connected to its maker, group number, and expiry date. Drug stores will be expected to output and record these identifiers upon buy and administer, making an unquestionable chain of care. This recognizability framework enables shoppers to utilize a versatile application to check

individual prescription units and immediately approve their realness, battling the flow of fake medications.

Public Straightforwardness and Strengthening: Increased transparency will help customers by allowing them to:

To find the best prices on medications, compare prices from various pharmacies.

Be made aware of potential cost gouging through the markup hailing framework.

Check the validness of their meds utilizing the versatile application

Introduction(with scope and outcomes):

To put it simply, because fake medicines are becoming more common and the pharmaceutical industry isn't very open about its practices, the government is starting a huge project. They are creating a special database to manage pharmacies. This project is meant to solve two main problems: stopping fake medicines from spreading and making the pharmaceutical industry more open and clear about what they do. It's a big step towards making sure people get safe and real medicines, while also making the industry more honest and transparent.

Unlike traditional systems that focus mainly on recipe and inventory management, this database project greatly expands its scope and ambitions. This database initiative focuses on even more public games involving questionable drug suppliers and unclear pricing mechanisms. This database initiative is a step forward in reforming the regulatory environment for drug distribution and control.

The SCOPE of this initiative exceeds the traditional database. Its mission is to include a comprehensive approach to the fight against counterfeit medicines and to increase the transparency of pricing mechanisms. Through the integration of advanced data analysis and real-time tracking functions, the database aims to provide a comprehensive platform to trace the origin of drugs, identify suspected suppliers, and illuminate hidden signs in the drug supply chain.

OUTCOME: This groundbreaking project aims to greatly decrease the circulation of fake medicines, make pricing information clearer, and make sure drug suppliers are more responsible. By promoting transparency and responsibility, this database is set to build more trust among consumers, healthcare workers, and regulators. This will bring about a new era of openness and trust in the pharmaceutical industry.

Project Objective(s):

The objective of our project is to develop a comprehensive pharmacy administration database managed by the government.

This database aims to address two main issues: combating the spread of counterfeit medicines and enhancing transparency within the pharmaceutical industry. By expanding the scope beyond traditional systems that focus on prescription and inventory management, this project seeks to tackle concerns related to questionable drug suppliers and unclear pricing practices.

The overarching goal is to create a platform that promotes transparency, accountability, and trust among consumers, healthcare professionals, and regulators by effectively managing drug distribution and control through advanced data analysis and real-time tracking capabilities.

A DISPLAY OF ALL THE TABLES THAT WERE GIVEN IN THE MOST BASIC AND SIMPLEST FORM SO AS TO GIVE A BETTER UNDERSTANDING

```
Pharmacy Table
+--------------+--------------+---------------------+
| pharmacy_id  | name         | location            |
+--------------+--------------+---------------------+
| PK           |              |                     |
| INT          | VARCHAR(255) | VARCHAR(255)        |
+--------------+--------------+---------------------+

Drug Table
+--------------+--------------+---------------------+
| drug_id      | name         | description         |
+--------------+--------------+---------------------+
| PK           |              |                     |
| INT          | VARCHAR(255) | TEXT                |
+--------------+--------------+---------------------+

Stock Table
+--------------+--------------+---------------------+
| pharmacy_id  | drug_id      | quantity            |
+--------------+--------------+---------------------+
| FK           | FK           | INT                 |
| INT          | INT          |                     |
+--------------+--------------+---------------------+
```

```
Price Table
+--------------+--------------+--------------------+
| pharmacy_id  | drug_id      | selling_price      |
|              |              | buying_price       |
|              |              | markup             |
+--------------+--------------+--------------------+
| FK           | FK           | DECIMAL(10,2)      |
| INT          | INT          | DECIMAL(10,2)      |
|              |              | DECIMAL(5,2)       |
+--------------+--------------+--------------------+

Transaction Table
+--------------+--------------+--------------------+
| transaction_id| pharmacy_id  | drug_id            |
|              | quantity     | date               |
|              | customer_name|                    |
+--------------+--------------+--------------------+
| PK           | FK           | FK                 |
| INT          | INT          | INT                |
| DATE         | VARCHAR(255) |                    |
+--------------+--------------+--------------------+
```

After normalization:
Implementation with MERN stack, (MongoDB, Express, Node.js, React)

Usage of MongoCompass as verification interface,
Integrated with UI using Bootstrap and Vite for React

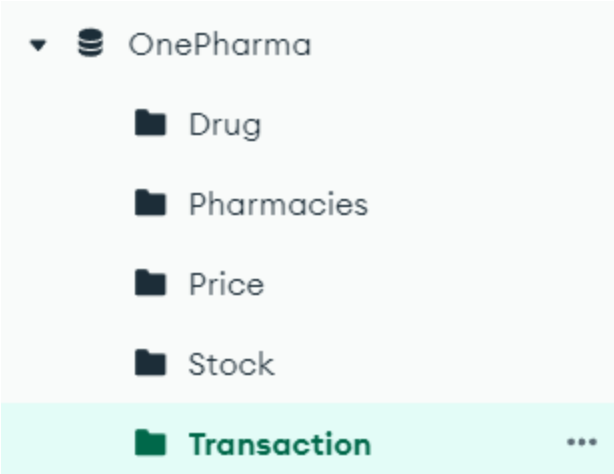
Packages used: in mongo-d

Mongoose

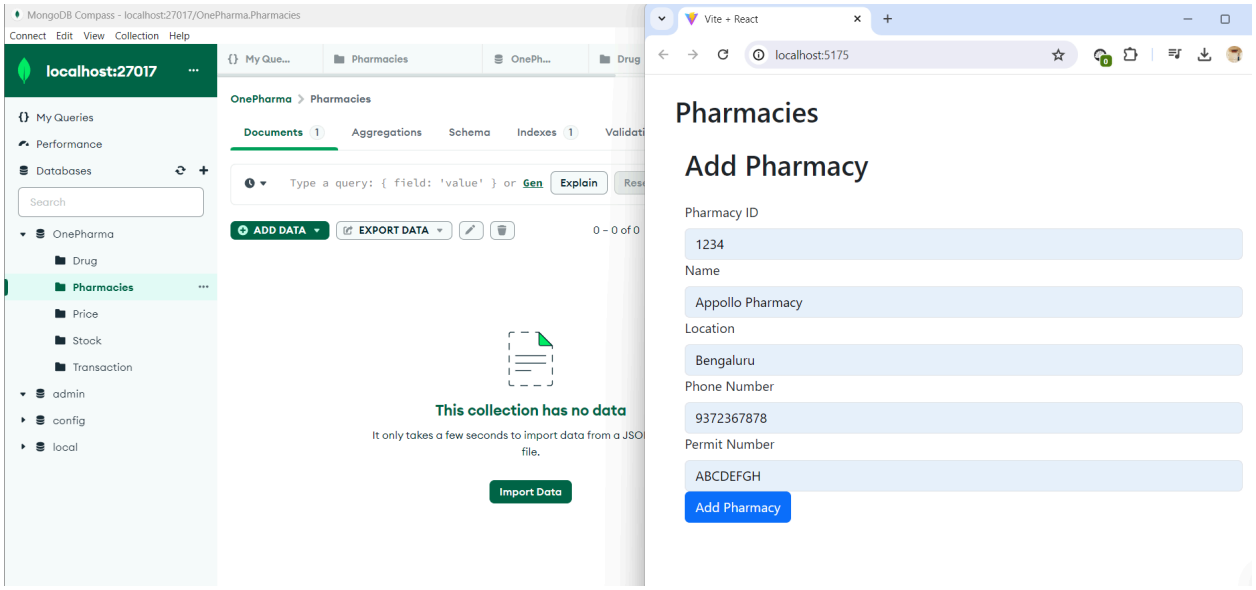
MongoShell

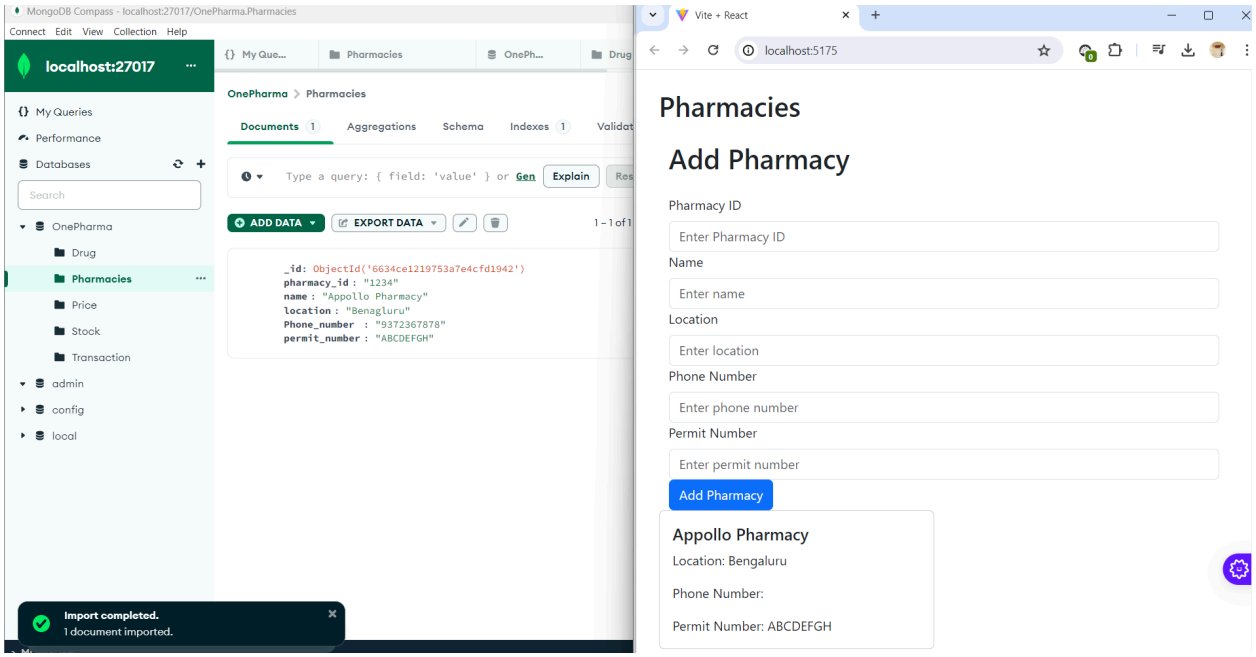
Express

Axios

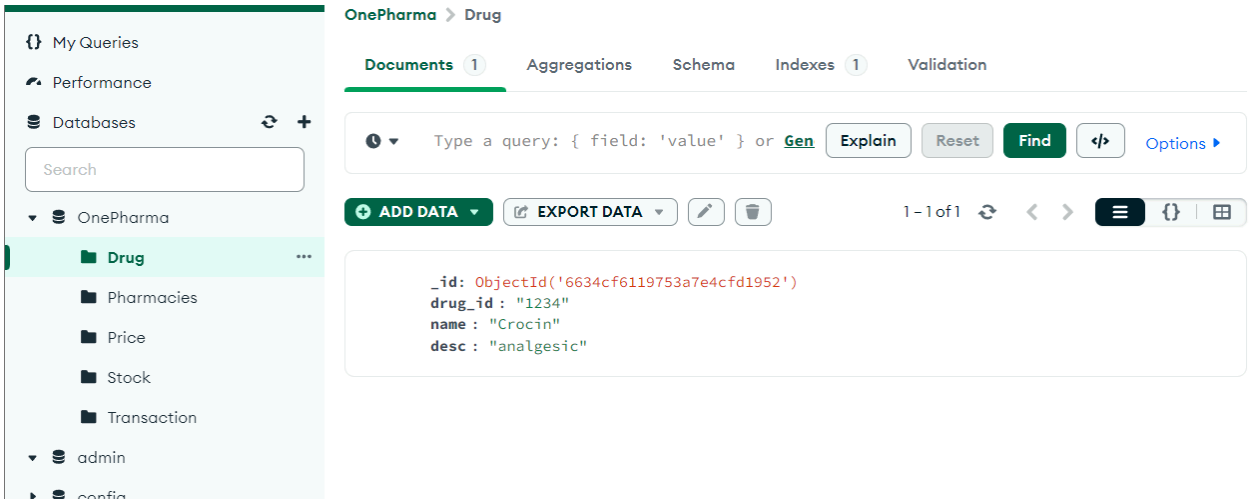


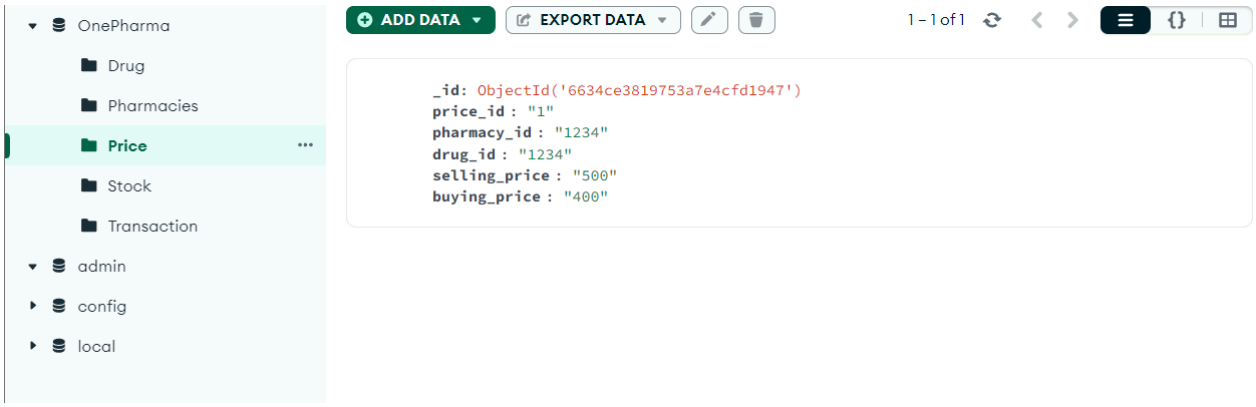
Real - time updation is as follows:

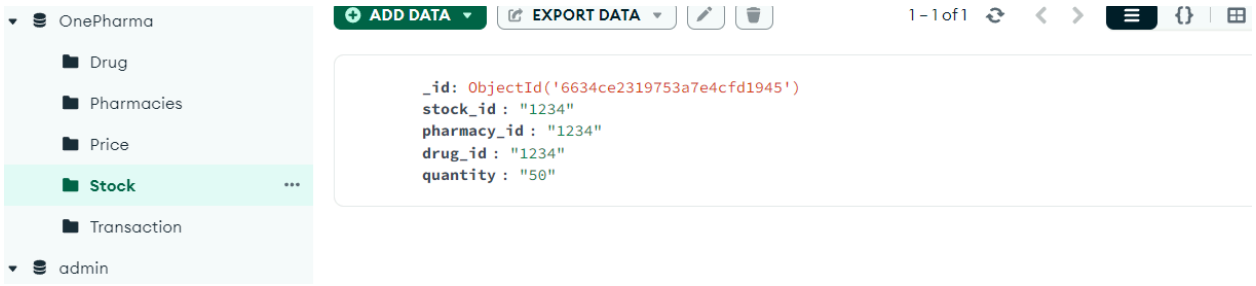


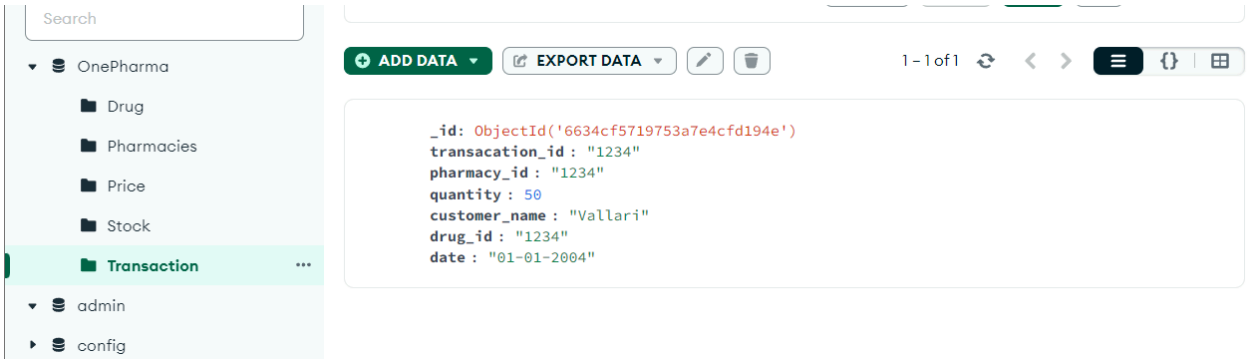


>>Database following sub-folders with input fields:









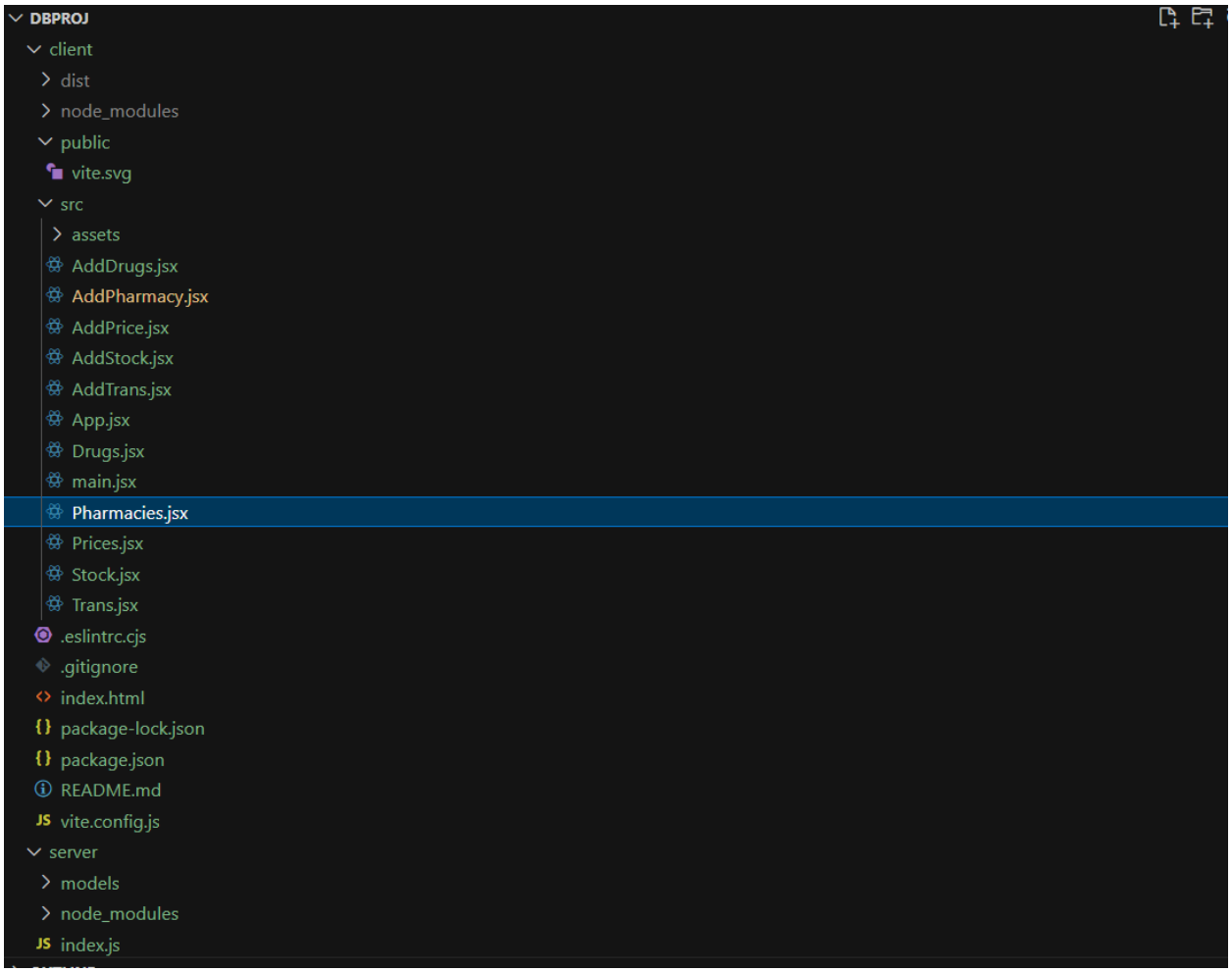
>> Integration code:

```
const express = require('express')
const mongoose = require('mongoose')
const cors = require('cors')
const PharmaciesModel = require("./models/Pharmacies")

const app = express()
app.use(cors())
app.use(express.json())

mongoose.connect("mongodb://127.0.0.1:27017/OnePharma")


app.listen(3001,()=> {
    console.log("Server is running")
})
```



File structure looks like this

CONCLUSION:

In conclusion, the successful implementation of a comprehensive pharmacy administration database managed by the government represents a significant stride forward in addressing two critical issues plaguing the healthcare system: the proliferation of counterfeit medications and the lack of transparency in pricing. By broadening the scope beyond mere prescription management or inventory control, this initiative demonstrates a commitment to safeguarding

public health and fostering greater accountability within the pharmaceutical industry. Through robust oversight and enhanced transparency measures, stakeholders can be assured of the authenticity and fair pricing of medications, thereby promoting trust and confidence in the healthcare system.